

Model Nesting Note (v1)

Canonical name: 05_Model_Nesting_Note_v1.md

Former name: `Model_Nesting_Note_v1.md` (renamed in place)

Date: 2026-06-24

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Audience: Manuscript §5 + staging #12 §3; Alex read-through; committee

Ladder terms: Rung / Path II / B-lite → [01_forward_plan_reading_guide.md](#) **Status:** Frozen for v1 manuscript — no new equations

1. Problem this note solves

The project had **one ontology** and **two readouts** that looked like two models:

- Charles's decomposable story: $L_{\text{net}} = B - D$ (development benefit minus competitive constraint)
- Alex's selection score: $S_i = A_i - \lambda L_{C, \text{LOO}}$ (ability minus congestion in selection)

This note states how they nest. **Path II is locked:** generative POC on pool mean; empirical stylized fact on LOO pool quality; predictions from decomposed empirical columns — **same object**, different rungs.

2. Structural object (one model)

$L_{\text{net}} = B(\cdot) - D(\cdot)$

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└┐ D: L00 congestion, crowding, finite slots, substitutability
└┐ B: visibility, norms, opportunity (minutes, elite exposure)

Symbol	Meaning
$B(\cdot)$	Developmental / signaling upside of stronger local peers
$D(\cdot)$	Competitive constraint / distinction compression from comparable peers
L_{net}	Net local environment effect on advancement propensity (reduced form)
A_i	Own ability / performance — outside L_{net} , enters selection directly

Alex score: $S_i = A_i - \lambda L_{C, \text{LOO}}$ is D entering the selection rule, jointly with own A_i . It is **not** the full reduced-form L_{net} and **not** a second mechanism.

3. Wang ladder (five rungs — v1)

Rung 1 Phenomenon
Realized pools $\rightarrow$ L00 pool quality (poolq_loo) $\rightarrow$ draft/promotion/tenure rate
 $\rightarrow$ inverted-U on L00 peer-quality proxy (Army , MBB , tenure  preliminary)

Rung 2 Minimal generative POC (basketball only)
Soft assignment $\rightarrow S_i = A_i - \lambda \cdot L_{\{C, L00\}} \rightarrow$ top-K selection
 $\rightarrow$ talent-only fails; congestion bends curves on **pool mean** axis
 $\rightarrow$ NOT bin-for-bin L00 pool-quality replication

Rung 2.5 Model-guided empirical features (PD12 P3)
poolq_loo (quality) vs crowding_smooth (congestion $C_{\{i, t\}}$)
 $\rightarrow$ theory proposes measurements, not only fits old curves

Rung 3 Predictions (named candidates – not all proven)
#1 Near-threshold heterogeneity (‘sports/538D_development.ipynb` CELL 4D  $\rightarrow$  ‘#08`)
#2 Peak shift with global Λ (Army-led prose hook)

Rung 4 Manuscript

4. Axis discipline (mandatory limitation sentence)

The empirical stylized fact is defined on **leave-one-out pool quality** among teammates; the minimal generative proof-of-concept conditions advancement on **whole-roster pool mean** and a selection score that penalizes **leave-one-out viable-peer congestion**. These are different conditioning objects: the generative exercise demonstrates that a congestion penalty in the selection rule can produce non-monotone advancement curves, while the empirical inverted-U remains the replicated fact on LOO pool quality — **not** a bin-for-bin reproduction of that fact by the current simulation knobs.

(Source: SCOUT model coherence §A3; frozen in D10 `score_equation_one_pager.md`.)

5. Mechanism column map (Rung 2.5 / 3)

Column	Leg	Role in v1
<code>poolq_loo</code> , <code>poolq_loo_mean</code>	Diagnostic of L_{net}	Rung 1 axis; mixes B–D in one neighbor statistic
<code>crowding_smooth</code> , viable-peer counts	D	PD12 congestion measurement
<code>peer_perf_sd_loo</code>	B–D diagnostic	Distinction vs crowding
<code>minutes</code>	B proxy (noisy)	Local opportunity
<code>perf</code> , own A_i	Outside L_{net}	Control / effect modifier

6. Cross-domain table (one row per setting)

Setting	Rung 1	Rung 2 generative	Rung 2.5	Rung 3
Army	CIF inverted-U on LOO pool minus mean	None (empirical leg)	Pool size / $\wedge$ proxies	Near-threshold; $\wedge$ peak-shift
Basketball	LOO <code>poolq_loo</code> inverted-U	Alex score POC (<code>sports/538D_development.ipynb</code> CELL 10)	Quality vs <code>crowding_smooth</code>	Near-threshold (<code>538D</code> CELL 4D)
Tenure	Stage 9 on <code>poolq_loo_mean</code> (preliminary)	None	<code>pool_size_oa_loo</code> optional	Deferred for v1

7. Explicitly deferred (stop rule)

Item	Why deferred
Multiplicative talent production function (Menger/Shockley)	Manuscript §5–§6 discussion; not v1 mechanism rewrite
Full generative $B(Q) - D(Q)$ decomposition	Not estimated in v1
LOO generative bin-for-bin match	Parallel north star; honest limitation
3-domain parametric identifiability (PD12 P1)	Post-draft
Fourth-domain falsification (PD12 P4)	Post-draft

8. Intellectual positioning (one paragraph)

Menger supplies selective ecology and assortative matching — who gets sorted into elite pools. **Barabási/Wang** supply the ladder from stylized fact to minimal mechanism to new measurements. **Alex (PD10–12)** supplies the operational score and the quality-vs-congestion split. **This project** performs the cross-domain empirical test and names the **development vs competitive constraint** distinction inside elite pools — the inverted-U as a local-competition signature, not a superstar-tail paper.

9. Artifacts (frozen)

Artifact	Path
D10 bundle	<code>datasets/mbb/exports_inverted_u_v0/scout_manuscript_v1/</code>
Axis table	<code>.../axis_table_generative_readouts.md</code>
Score one-pager	<code>.../score_equation_one_pager.md</code>
Tenure inference sample	<code>tenure_pipeline/faculty_panel_inference_v1.csv</code>
Claim discipline	<code>3-Master_Plan/07_Claim_language_guardrails.md</code>

End Model Nesting Note v1.